

# Lab 4

## Question 1

```
Tilix: thecoon@thecoon: ~  
1: thecoon@thecoon: ~  
thecoon@thecoon:~$ apt search "tetris clone"  
bastet/stable 0.43.2-2 amd64  
  ncurses Tetris clone with a bastard algorithm  
  
ttris/stable 2.0.3-1 amd64  
  very polished Tetris clone with CPU opponents  
  
tint/stable 0.07 amd64  
  Tetris clone for text based terminal  
  
thecoon@thecoon:~$ apt search "tetris"  
bastet/stable 0.43.2-2 amd64  
  ncurses Tetris clone with a bastard algorithm  
  
blockattack/stable 2.9.0-1+b1 amd64  
  puzzle game inspired by Tetris  
  
blockout7/stable 2.5+dfsg1-2 amd64  
  Tetris like game (3D-tetris)  
  
blocks-of-the-undead/stable 1.0-7 amd64  
  Tetris Attack clone with spooky undertones  
  
blocks-of-the-undead-data/stable 1.0-7 all  
  Tetris Attack clone with spooky undertones - data files  
  
badgames/stable 2.17-35 amd64  
  collection of classic textual unix games  
  
crack-attack/stable 1.1.14-11 amd64  
  multiplayer OpenGL puzzle game like "Tetris Attack"  
  
cuvo/stable 2.1.0-1 amd64  
  Tetris-like game with very impressive effects  
  ESCOD  
  
petris/stable 1.0.1-11.1 amd64  
  Peter's Tetris - a Tetris(TM) clone  
  
quadrappel/stable 1:40.2-4 amd64  
  popular Russian game, similar to Tetris  
  
stax/stable 1.37-2+b2 amd64  
  collection of puzzle games similar to Tetris Attack  
  
tetrinet-client/stable 0.11+CVS20070911-2.1 amd64  
  textmode client for tetrinet, a multiplayer tetris-like game  
  
tetrinet-server/stable 0.11+CVS20070911-2.1 amd64  
  server for tetrinet, a multiplayer tetris-like game  
  
tetrinetx/stable 1.13.16-18 amd64  
  game server for Tetrinet  
  
tint/stable 0.07 amd64  
  Tetris clone for text based terminal  
  
vitetris/stable 0.59.1-3 amd64  
  Virtual terminal *tris clone  
  
xwelltris/stable 1.0.1-18 amd64  
  3D Tetris like popular game similar to Welltris  
  
thecoon@thecoon:~$ apt search --names-only "tetris"  
games-tetris/stable 5 all  
  Debian's tetris-like games  
  
vitetris/stable 0.59.1-3 amd64  
  Virtual terminal *tris clone  
  
thecoon@thecoon:~$
```

```

1/1 v + [ ] [ ]
Tilix: thecoon@thecoon: ~
thecoon@thecoon:~$ sudo apt install blockout2+ cuyo+ blockattack-
The following packages were automatically installed and are no longer required:
  libfat10 libopusfile0 libphysfs1 libSDL2-image-2.0-0 libSDL2-mixer-2.0-0 libSDL2-ttf-2.0-0 libxmp4
Use 'sudo apt autoremove' to remove them.

Installing:
  blockout2 cuyo

Installing dependencies:
  cuyo-data libmikmod3 libSDL-mixer1.2 libSDL1.2debian

REMOVING:
  blockattack

Summary:
  Upgrading: 0, Installing: 6, Removing: 1, Not Upgrading: 6
  Download size: 8,078 kB
  Space needed: 4,102 kB / 27.5 GB available

Continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 libmikmod3 amd64 3.3.13-1 [135 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libSDL1.2debian amd64 1.2.68-3 [85.9 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libSDL-mixer1.2 amd64 1.2.12-19+b1 [81.6 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 blockout2 amd64 2.5+dfsg1-2 [4,228 kB]
Get:5 http://deb.debian.org/debian trixie/main amd64 cuyo-data all 2.1.0-1 [3,372 kB]
Get:6 http://deb.debian.org/debian trixie/main amd64 cuyo amd64 2.1.0-1 [175 kB]
Fetched 8,078 kB in 1s (10.9 MB/s)
(Reading database ... 195880 files and directories currently installed.)
Removing blockattack (2.9.0-1+b1) ...
Selecting previously unselected package libmikmod3:amd64.
(Reading database ... 195853 files and directories currently installed.)
Preparing to unpack .../0-libmikmod3_3.3.13-1_amd64.deb ...
Unpacking libmikmod3:amd64 (3.3.13-1) ...
Selecting previously unselected package libSDL1.2debian:amd64.
Preparing to unpack .../1-libSDL1.2debian_1.2.68-3_amd64.deb ...
Unpacking libSDL1.2debian:amd64 (1.2.68-3) ...
Setting up libmikmod3:amd64 (3.3.13-1) ...
Setting up libSDL1.2debian:amd64 (1.2.68-3) ...
Setting up cuyo-data (2.1.0-1) ...
Setting up cuyo (2.1.0-1) ...
Setting up blockout2 (2.5+dfsg1-2) ...

thecoon@thecoon:~$ bastet
thecoon@thecoon:~$ man bastet
thecoon@thecoon:~$ sudo apt remove bastet
The following package was automatically installed and is no longer required:
  libboost-program-options1.83.0
Use 'sudo apt autoremove' to remove it.

REMOVING:
  bastet

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
  Freed space: 245 kB

Continue? [Y/n] y
(Reading database ... 195802 files and directories currently installed.)
Removing bastet (0.43.2-2) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for gnome-menus (3.36.0-3) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for desktop-file-utils (0.28-1) ...
thecoon@thecoon:~$ sudo apt purge bastet
The following package was automatically installed and is no longer required:
  libboost-program-options1.83.0
Use 'sudo apt autoremove' to remove it.

REMOVING:
  bastet+

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
  Space needed: 0 B / 27.8 GB available

Continue? [Y/n] y
(Reading database ... 195794 files and directories currently installed.)
Removing bastet (0.43.2-2) ...

```

```

Tilix: thecoon@thecoon: ~
thecoon@thecoon:~$ apt search "tetris clone"
bastet/stable 0.43.2-2 amd64
ncurses Tetris clone with a bastard algorithm

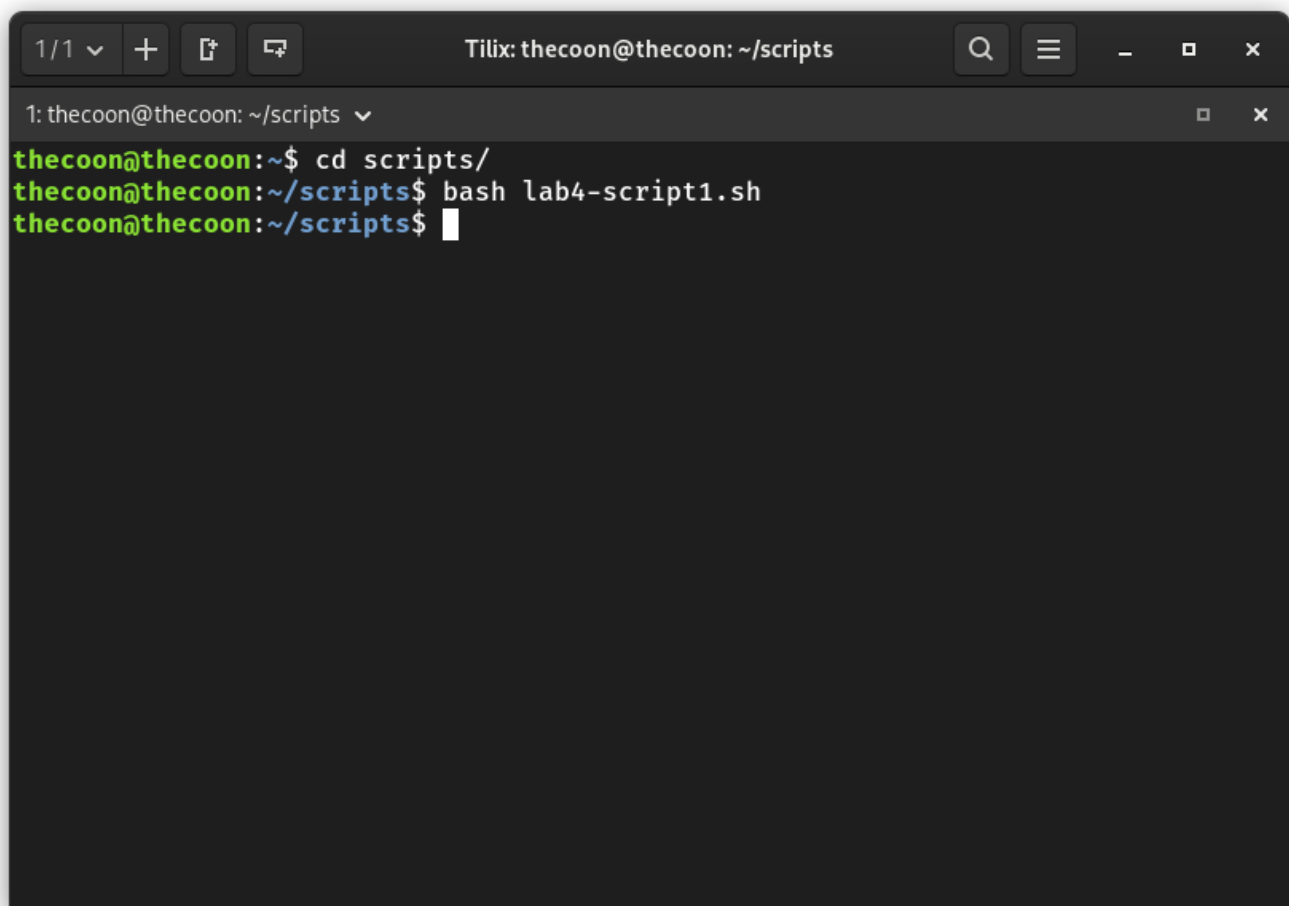
ltris/stable 2.0.3-1 amd64
very polished Tetris clone with CPU opponents

tint/stable 0.07 amd64
Tetris clone for text based terminal

thecoon@thecoon:~$

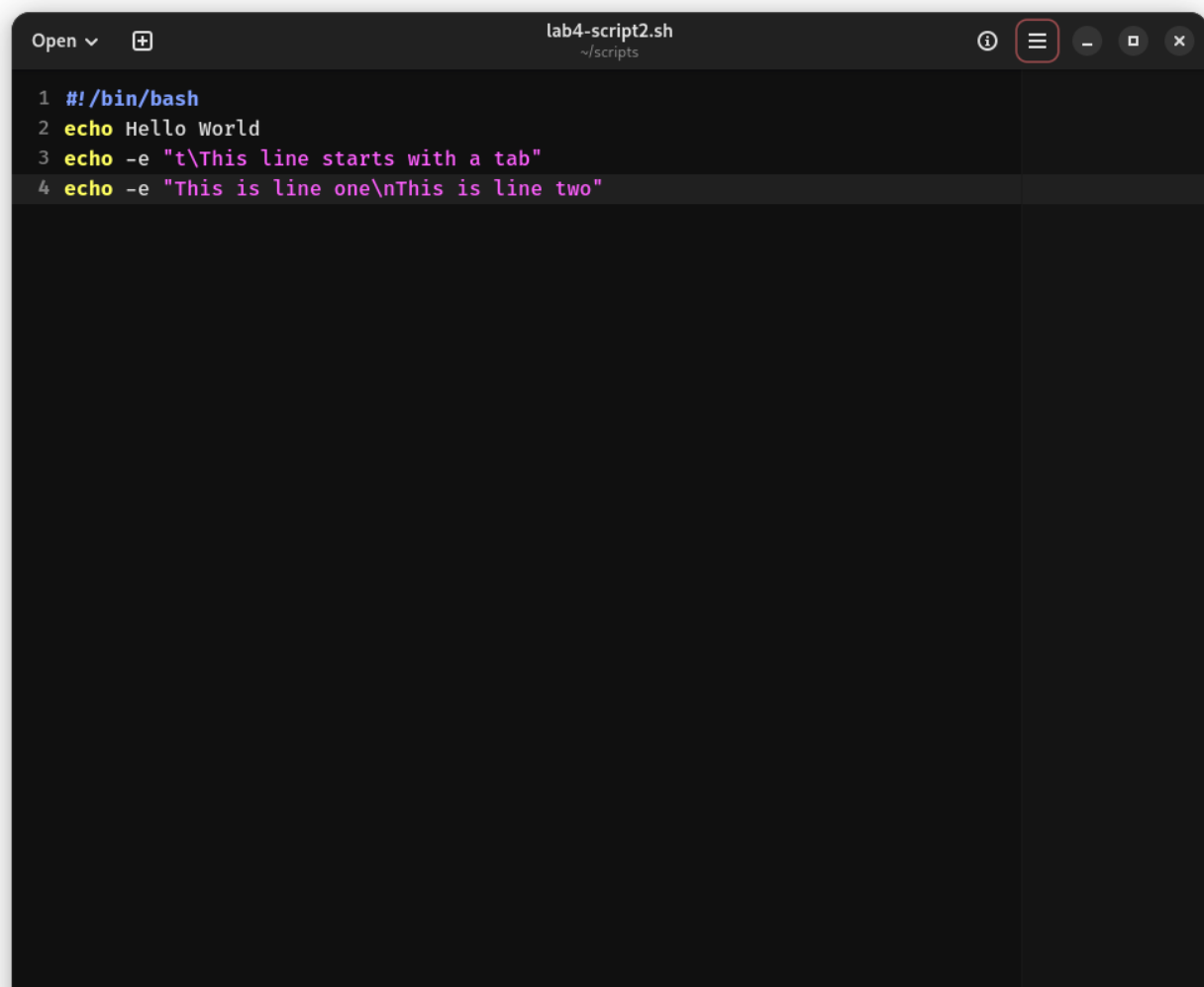
```

## Question 2



A terminal window titled 'Tilix: thecoon@thecoon: ~/scripts'. The window shows a sequence of commands: 'cd scripts/' followed by 'bash lab4-script1.sh'. The prompt changes from '~\$' to '~/scripts\$' after the first command. The second command is entered but not yet executed, as indicated by the cursor at the end of the line.

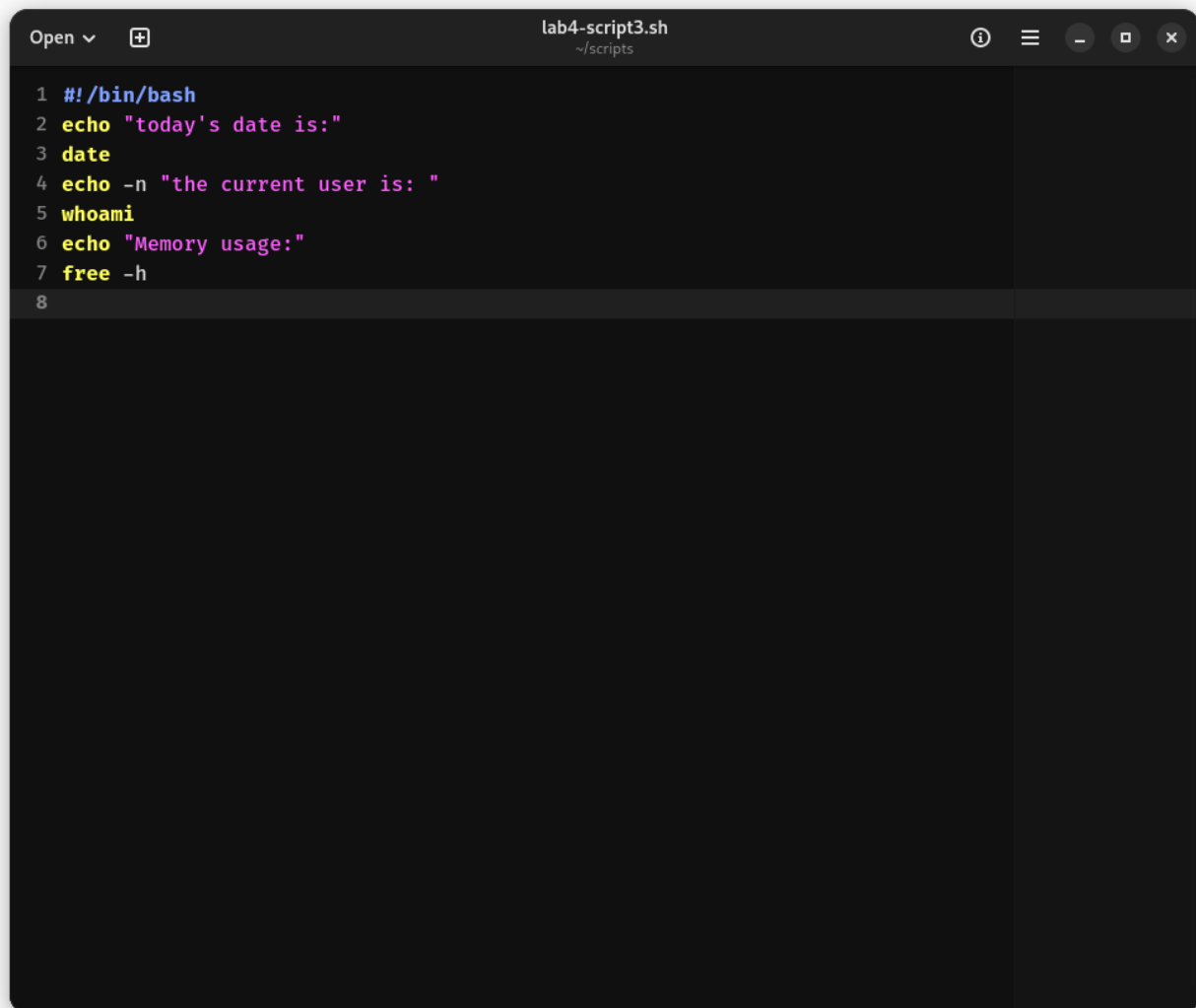
```
1: thecoon@thecoon: ~/scripts
thecoon@thecoon:~$ cd scripts/
thecoon@thecoon:~/scripts$ bash lab4-script1.sh
thecoon@thecoon:~/scripts$
```



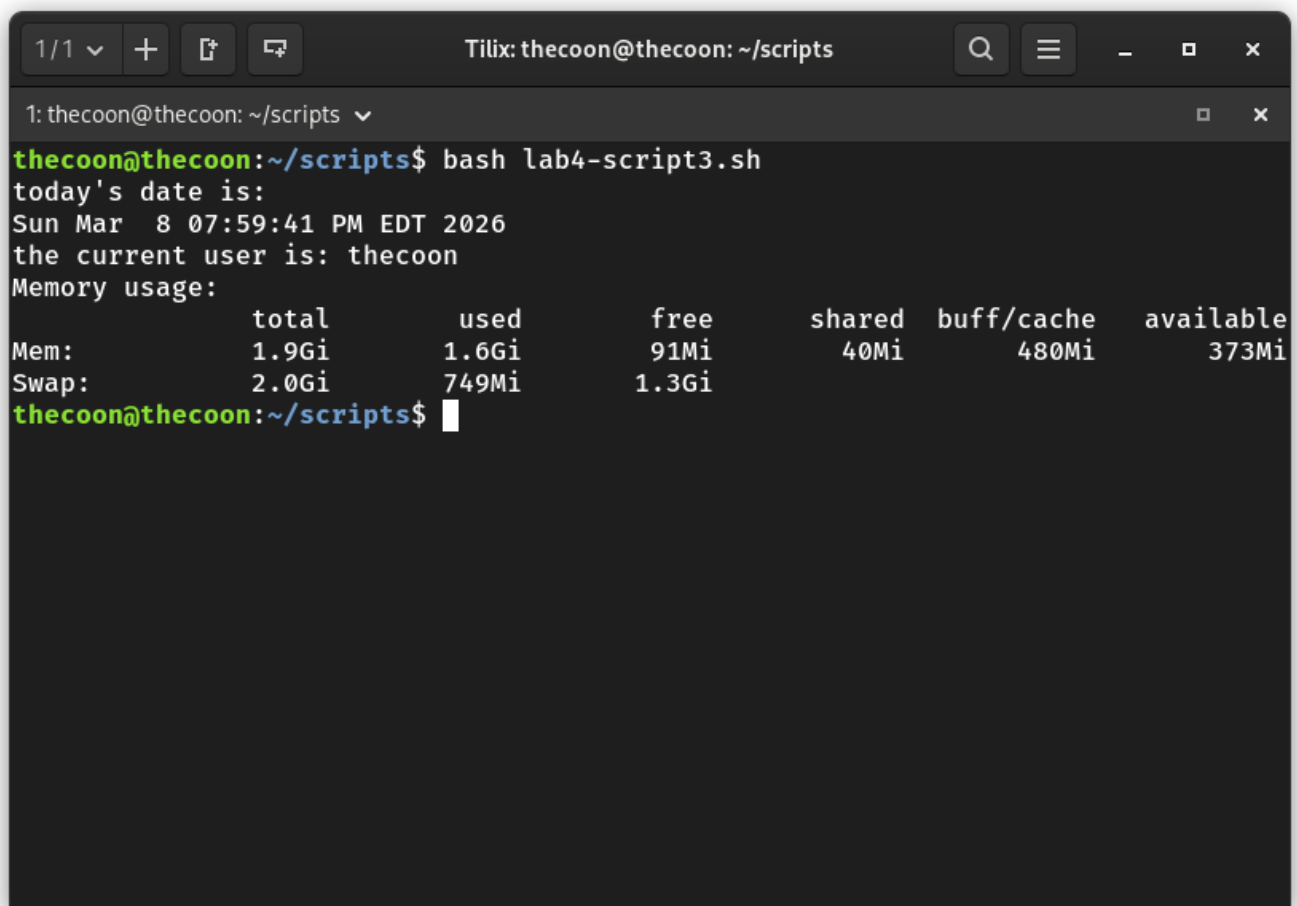
An editor window titled 'lab4-script2.sh' with the subtitle '~/scripts'. The window contains four lines of shell script code. The third line is highlighted in grey. The menu bar includes 'Open', a plus icon, and standard window controls. The menu icon (three horizontal lines) is highlighted with a red rectangle.

```
1 #!/bin/bash
2 echo Hello World
3 echo -e "t\\This line starts with a tab"
4 echo -e "This is line one\\nThis is line two"
```



A screenshot of a terminal window with a dark background. The window title bar shows 'lab4-script3.sh' and '~/.scripts'. The script content is as follows:

```
1 #!/bin/bash
2 echo "today's date is:"
3 date
4 echo -n "the current user is: "
5 whoami
6 echo "Memory usage:"
7 free -h
8
```



A terminal window titled 'Tilix: thecoon@thecoon: ~/scripts' showing the execution of a script. The prompt is 'thecoon@thecoon:~/scripts\$'. The script 'lab4-script3.sh' is run, displaying the current date and user. It then shows memory usage with a table of values.

```
1: thecoon@thecoon: ~/scripts ▾
thecoon@thecoon:~/scripts$ bash lab4-script3.sh
today's date is:
Sun Mar  8 07:59:41 PM EDT 2026
the current user is: thecoon
Memory usage:

```

|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 1.9Gi | 1.6Gi | 91Mi  | 40Mi   | 480Mi      | 373Mi     |
| Swap: | 2.0Gi | 749Mi | 1.3Gi |        |            |           |

```
thecoon@thecoon:~/scripts$
```

## Challenge Question

---

Open + challenge\_lab4.sh ~/scripts

lab4-script3.sh challenge\_lab4.sh

```
1 #!/bin/bash
2 echo "_____ "
3 figlet SYSTEM
4 echo -n "DATE UTC: "
5 date -u
6 echo -n "HOSTNAME: "
7 hostname
8 echo -n "UPTIME: "
9 uptime -p
10 echo -n "USER: "
11 whoami
12 figlet MEMORY
13 free -h
14 figlet DISK
15 df -h
16 echo "_____ "
17
18
```

1/1 + Tili: thecoon@thecoon: ~/scripts

1: thecoon@thecoon: ~/scripts

thecoon@thecoon:~/scripts\$ bash challenge\_lab4.sh

```
SYSTEM
DATE UTC: Mon Mar 9 12:27:16 AM UTC 2026
HOSTNAME: thecoon
UPTIME: up 7 hours, 18 minutes
USER: thecoon

MEMORY
Mem:      total      used      free      shared  buff/cache  available
Swap:    2.0Gi      882Mi      1.1Gi           38Mi      499Mi      401Mi

DISK
Filesystem      Size  Used Avail Use% Mounted on
udev            958M   0 958M   0% /dev
tmpfs           198M  4.3M  194M   3% /run
/dev/sda1       38G   9.6G   26G  28% /
tmpfs           987M  45M  943M   5% /dev/shm
tmpfs           5.0M   0 5.0M   0% /run/lock
tmpfs           1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs          987M  4.0M  983M   1% /tmp
tmpfs          198M  192K  198M   1% /run/user/1000
```